03 · Case L, question 5: otosclerosis

The words of this track, as it says them. Bold brown is the examiner.

Case L, question five. Otosclerosis. Four marks.

A CT scan of Jack's ears shows no signs of otosclerosis. What kind of deafness would otosclerosis have caused? What specific structure of the ear is damaged by that condition, in what part of the ear is it found, and how does that damage affect hearing?

This is the flip side of Jack's case. His deafness is sensorineural. Otosclerosis would be the other kind. So really, this question is asking: do you know the difference?

The hook: ossicles stuck, no vibration, conduction deafness.

What kind of deafness? Conduction deafness. Something stops sound being conducted to the inner ear. Her deck lists otosclerosis right alongside earwax, a perforated eardrum, and middle ear infections.

Which structure? The ossicles: the three tiny ear bones, the malleus, the incus and the stapes. Her deck calls it sclerosis of the ossicles, and Patton says the irregularities are in the stapes. So name the ossicles, and for the extra detail, the stapes above all.

Where are they? In the middle ear. Their job is to carry the vibration from the eardrum, the tympanic membrane, across to the oval window. The stapes is the one that pushes on the oval window.

How does that affect hearing? In otosclerosis, the ossicles become attached to the walls of the middle ear. That's her wording, from her ear learning page, and it's worth writing word for word: the ossicles become attached to the walls of the middle ear, which interferes with the vibrations. Stuck bones can't vibrate, so they can't amplify the sound onto the oval window, and the sound never properly reaches the inner ear.

And one more line, for the comparison: the inner ear and its hair cells are fine, so sound can still be conducted through the bones of the skull. That's the difference from Jack, whose hair cells are broken.

The traps. Otosclerosis is conduction deafness, not sensorineural. Her multiple choice key comes at it from the other side: sensorineural deafness can come from everything except damage to the ossicles. Second: name the specific structure. "The middle ear" is a place, not a structure. The structure is the ossicles. Third: keep your ears straight. Ossicles, middle ear. Organ of Corti and hair cells, inner ear.

Once more. Otosclerosis: what kind of deafness, what's damaged, where, and how it affects hearing.

Conduction deafness. The ossicles, the malleus, incus and stapes, are damaged. They sit in the middle ear and carry vibration from the eardrum to the oval window. In otosclerosis they

become attached to the walls of the middle ear, so they can't vibrate or amplify the sound, and it isn't conducted to the inner ear. The inner ear and its hair cells are fine, which is the difference from Jack's sensorineural loss.